What scaling up RHD screening and prophylaxis might buy in Indonesia and Uganda: preliminary results from a first-diagnosis model

Technical review of the rheumatic heart disease investment-case model

William Garcia

University of Washington · World Bank

November 2026

These results are preliminary: the first-diagnosis model passes its structural checks but fails its fit checkpoint in both countries

Checkpoint	Indonesia	Uganda
Structural validity (accounting, D-only, one-time diagnoses)	PASS	PASS
Numerical validity (starts converge; 2× burn-in)	PASS	PASS
Epidemiological fit, every scored year 2010–2019	**FAIL**	**FAIL**
Future plausibility (2026 baseline)	**FAIL**: deaths 1.0× GBD; D share 10%	**FAIL**: deaths 4.8× GBD; D share 12%
Baseline label saved with the run	INADEQUATE	INADEQUATE

Statuses read from the saved calibration and model outputs. Model update (6 Nov 2026): adult B→C and C→D hazards fixed at 0.10 (IDN) / 0.50 (UGA)× the childhood hazards; SAP: assumed 55% progression-hazard reduction (M = 0.45, ages ≤ 39) replaces the GOAL child HRs; 9 Nov: fixed D death by age (< 20 / ≥ 20); both countries recalibrated and re-run. 05–08 were run with the fit outside tolerance by explicit user decision. Confirmatory echo is not costed. Tolerances: |log ratio| < 0.15 (prevalence), 0.25 (deaths).

We ask what scaling up echo screening and prophylaxis would buy, and at what cost

1. **Health:** how many rheumatic heart disease (RHD) deaths does scale-up avert in Indonesia and Uganda, 2026–2100?
2. **Services:** how many screens, confirmatory echoes, first diagnoses and prophylaxis person-years does it add?
3. **Value:** does the monetised benefit exceed the incremental cost?

Scope: national, both sexes, ages 0–100+; secondary prevention (screening, confirmation, benzathine penicillin G (BPG) prophylaxis) only.

About 2.2 million people live with RHD in the two countries, and most are undiagnosed in the model

2.16 million

people living with RHD in Indonesia and Uganda, 2023

Modelled share of classifiable RHD with a confirmed diagnosis entering 2026: 27% in Indonesia, 19% in Uganda; on BPG 14% and 8%.

GBD 2023 RHD prevalence, published counts summed over age and sex (Indonesia 1,400,743; Uganda 762,663). Model shares are end-of-2026 reference-arm stocks, grown from provisional initial care shares. Amber marks provisional assumptions throughout.

The model builds on an established RHD investment case, but its structure and parameters are provisional

Builds on

- Investment-case design of Coates et al. (2021): in the African Union, secondary prevention with secondary and tertiary care returned a benefit–cost ratio (BCR) of 4.7 to 2030
- World Heart Federation (WHF) echo stages A–D (Rwebembera et al., 2023)
- GBD 2023 burden; UN World Population Prospects (WPP) 2024

Boundary

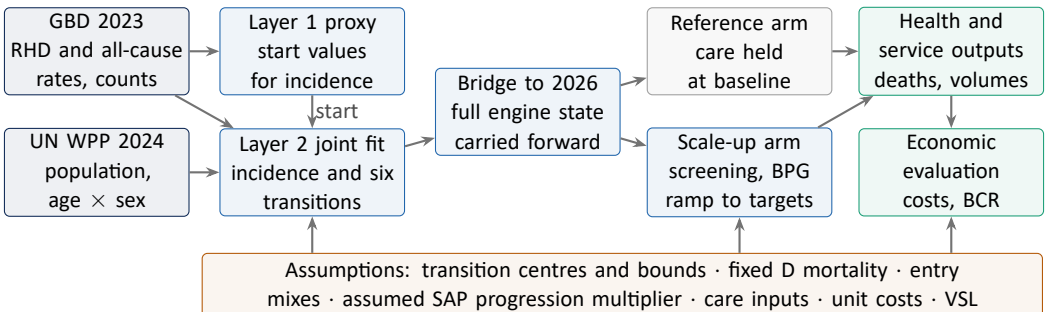
Deterministic point estimates; no sensitivity analysis.

Disease history and first confirmed diagnosis are separated; stage-D-only mortality and a progression effect of SAP are provisional choices.

The fit fails its tolerances; results are **not** a credible baseline yet.

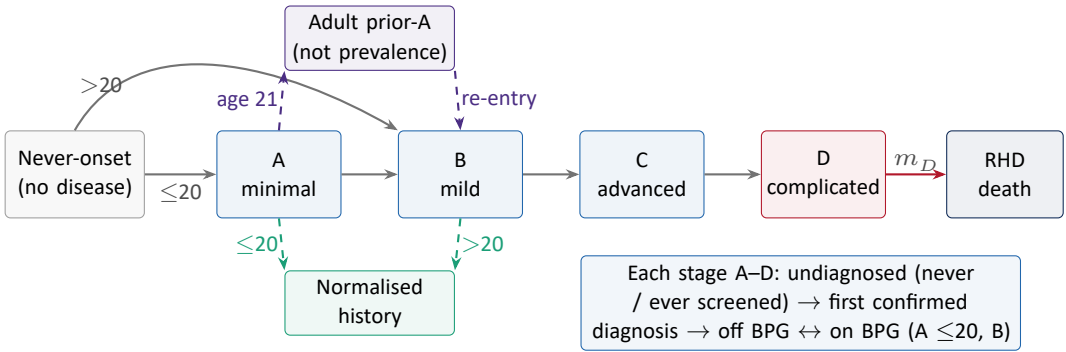
How the model works

Four linked steps turn GBD and WPP inputs into health and economic contrasts



GBD supplies the targets; WPP the population. Layer 2, the bridge and both arms run one annual engine (abcd_first_diagnosis_v1_adult_prog_mult_sap55_dmort_age). SAP: secondary antibiotic prophylaxis. VSL: value of a statistical life.

New disease enters mainly A before age 21 and B after; screening finds it, and only stage D dies of RHD



Regression C→B and B→A at ages ≤20; other-cause death from every state; D→C fixed at 0

New disease enters A 75%/B 25% at attained ages ≤ 20 and B above; the boundary is a modelling convention, not a WHF rule. Surgery is a service on C→D and D death, not a state.

Every stage is split by care status, and only some stocks count as prevalence or receive SAP

Stock	Stages	Prevalence	SAP	Screenable
Undiagnosed, never screened	A–D	Yes	No	Yes
Undiagnosed, screened (missed / unconfirmed)	A–D	Yes	No	No (one screen)
Confirmed, not on BPG	A–D	Yes	Eligible if A ($\leq$20) or B	No
Confirmed, on BPG	A ($\leq$20), B	Yes	Yes	No
Adult prior-A (age $>$20)	–	No	No	If never screened
Normalised history	–	No	No	If never screened
Never-onset	–	No	No	If never screened

Classifiable RHD (GBD prevalence mapping) = undiagnosed + diagnosed in A–D; RHD deaths come from diagnosed and undiagnosed D. A first confirmed diagnosis happens once; nobody returns to undiagnosed.

Six childhood transitions are calibrated; adult B→C and C→D run at 0.10 (IDN) / 0.50 (UGA) times the childhood hazard

Annual probability	Age ≤20	Age >20	Indonesia	Uganda
A → normalised	calibrated	0	0.249	0.080
A → B (adult: prior-A → B)	calibrated	same value	0.072	0.030
B → A	calibrated	0	0.051	0.080
B → C (child / adult)	calibrated	hazard × m	0.125 / 0.013	0.080 / 0.041
C → B	calibrated	0	0.034	0.025
C → D (child / adult)	calibrated	hazard × m	0.250 / 0.028	0.080 / 0.041
Adult progression hazard multiplier m	–	fixed	0.10	0.50
B → normalised	0	fixed	0.020	0.020
D → RHD death (age <20 / ≥20)	fixed	fixed	0.160 / 0.040	0.060 / 0.030

Values in force (calibrated or fixed). Adult B→C and C→D: $p_{\text{adult}} = 1 - (1 - p_{\text{child}})^m$, every disease and diagnosis history; age 20 uses the childhood values. Annual probabilities are converted once to hazards. A→C, A/B/C RHD death and D→C are exact zeros. D death depends only on attained age (age 20 takes the adult value); the late-entry D-death multiplier is removed (value 1).

At age 21, surviving A becomes prior-A: about 5,300 people a year in Indonesia, 7,600 in Uganda

Reference arm, 2026	Indonesia	Uganda
New modelled disease (A $\leq$20, B $>$20)	47,118	35,228
A reclassified to prior-A at 21	5,293	7,579
Prior-A re-entering B	5,054	3,491
Adult prior-A stock (end of year)	64,746	112,514
Normalised history stock (end of year)	1,034,615	294,989

New disease is not diagnosis. Re-entry from prior-A is tracked separately from first entry and from first diagnosis.

Competing exits are split on the hazard scale, so every stock balances exactly

Each state faces several annual risks at once; converting them to hazards keeps every outflow non-negative.

$$p_{s\to e} = \left(1 - e^{-H_s}\right) \frac{h_e}{H_s}, \qquad h_e = -\ln\left(1 - p_e\right), \qquad H_s = \sum_{e \in \mathcal{E}_s} h_e$$

State s keeps the share e^{-H_s}. For stage B at age > 20: to C, to normalised, other-cause death (its RHD-death hazard is exactly zero). All probabilities are per year.

Each year reconciles the population, updates care, applies exits, adds new disease, then ages everyone

1. Stocks enter year t already aged; never-onset and normalised pools absorb WPP births and migration
2. Care update: screening, confirmation, clinical detection, BPG retention and initiation
3. Competing exits split on the hazard scale; on-BPG people get the SAP hazard multipliers
4. New modelled disease from the never-onset pool joins A (≤ 20) or B (> 20) at year end
5. Survivors age one year; surviving A at 20 becomes prior-A; 100+ is open

Disease and prior-A stocks are never rescaled. New disease first faces stage hazards the next year. Prevalence is observed as the mean of the start and end stocks.

A first confirmed diagnosis needs a screen, a true positive and a completed confirmation — or clinical detection

$$\mathrm{dx}_t^{\mathrm{scr}} = q_t \, s \, \kappa \, \phi \, U_t^{\mathrm{ns}}, \qquad q_t = 1 - (1 - c_t)^{1/35}, \qquad \mathrm{dx}_t^{\mathrm{clin}} = k \, (U_t - \mathrm{dx}_t^{\mathrm{scr}})$$

U^{ns}: undiagnosed, never screened; c_t: probability of one screen across ages 5–39; sensitivity s = 0.90, confirmation completion κ = 0.90, confirmation sensitivity ϕ = 1.00; clinical detection k = 0.02. Screening encounters $= q_t \times$ never-screened people without a prior diagnosis.

Perfect specificity assumed (no false positives). Window target c: reference 5% (Indonesia), 1% (Uganda); scale-up 80% by 2050 (q = 0.045 a year).

SAP is assumed to cut A$\rightarrow$B and B$\rightarrow$C hazards by 55% on BPG below age 40: an expert-informed scenario, not an estimate

	Reference	Scale-up (2050)
Eligibility (unchanged)	A ($\leq$20), B; not C/D	same
BPG initiation (diagnosed, off BPG)	50%	90%
BPG retention (annual)	90%	95%
Adherence: good / partial share	60% / 20%	85% / 10%
Protection factor a (good + 0.5 $\times$ partial)	0.70	0.90
Assumed multiplier M (full adherence), A$\rightarrow$B and B$\rightarrow$C, age $\leq$39	0.45	same
On-BPG multiplier $1 - a(1 - M)$, age $\leq$39	0.615	0.505
Age $\geq$40; C$\rightarrow$D; D death	1 (no modelled effect)	same

Default: assumed 55% reduction in progression hazards (expert-informed scenario), informed by evidence on prevention of recurrent rheumatic fever. Manyemba et al. (Cochrane, PMID 12137650): RR 0.45 for recurrent RF (55% RRR), one trial; not a HR, not valve progression; M assumed. $h = h_0[1 - a(1 - M)]$; illustrative a = 0.90: 0.505. GOAL child HRs: documented alternative.

Observed data anchor demography and burden; stage dynamics and care are calibrated or assumed

Input	Source	Status
Population	UN WPP 2024	Observed
Prevalence, deaths	GBD rate $\times$ WPP	Observed
Background deaths	GBD all-cause $-$ RHD	Observed
RHD incidence	GBD $\times$ Layer-2 α	Calibrated
Six transitions	Layer 2 within bounds	Partly identified
Stage-D mortality	$<$20 / 20+: 0.16 / 0.04 (IDN); 0.06 / 0.03 (UGA)	Fixed, provisional
Entry mixes, adult normalisation	Modelling assumption	Fixed, new
Initial diagnosed / on-BPG share	5% / 4% (Indonesia)	Provisional
Incidence trend	Assumption	Provisional

Costs price screening encounters and BPG person-years; confirmation is not priced

Input (2025 US$)	Indonesia	Uganda	Status
Screening encounter	US$1.10	US$0.52	Undocumented
Confirmatory echo	not costed	not costed	Omitted
BPG (SAP) person-year	US$110	US$52	Undocumented
Valve surgery	US$8,998	US$6,560	Undocumented
GNI per capita, PPP, 2019	int$11,620	int$2,380	World Bank WDI, 2019
VSL in 2026; discount rate	US$373,000; 3%	US$47,000; 3%	Reference case

Volumes come from the engine: encounters, on-BPG person-years (A ≤ 20 and B), surgeries. Unit-cost sources are not recorded; price year assumed 2019, converted to constant 2025 US$ with each country's CPI and exchange rates (factor 1.000 Indonesia, 1.312 Uganda). Total cost is understated by the confirmation cost.

How the model was calibrated

Layer 2 fits prevalence 2.5 times as heavily as deaths, with child and adult components weighted equally

Layer 1: aggregate proxy (2000–2019)

- Well–sick–dead proxy; GBD counts
- Weights incidence / prevalence / deaths 1 / 2.5 / 1
- Role: start values only

Layer 2: production engine (scored 2010–2019)

- Prevalence 2.5, deaths 1.0, incidence 0.0
- Each split 0–19 / 20+ at 0.5 / 0.5
- Free: 6 childhood transitions + 12 incidence α
- Adult B$\rightarrow$C, C$\rightarrow$D: hazard $\times$ 0.10 (IDN) / 0.50 (UGA) (fixed)
- Burn-in 120 cycles, per-capita balanced

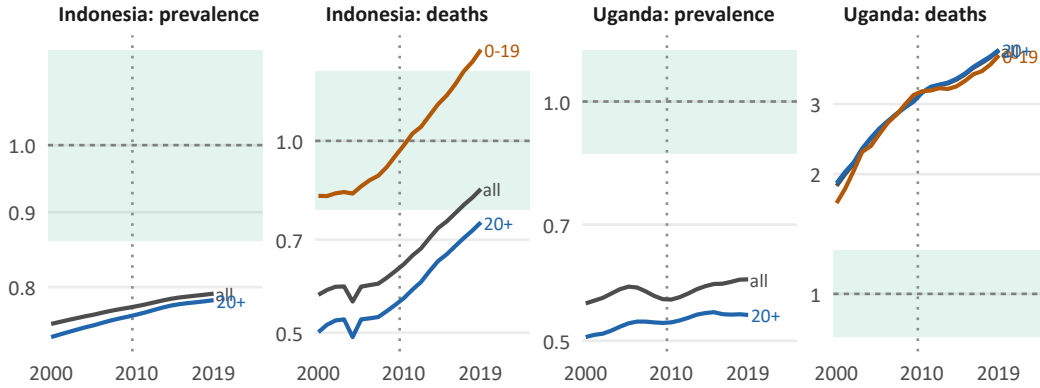
Targets: GBD rate $\times$ WPP population; prevalence = classifiable RHD (diagnosed + undiagnosed), prior-A and normalised excluded. Prior weight λ = 0.05.

Deaths are 0.64–0.84 times GBD in Indonesia and 3.21–4.09 in Uganda across 2010–2019; no checkpoint quantity is in tolerance in every year

	Indonesia		Uganda	
Model / GBD target	Ratio range	Yrs in tol.	Ratio range	Yrs in tol.
Prevalence, all ages	0.78–0.79	0/10	0.56–0.60	0/10
Prevalence, 20+	0.77–0.78	0/10	0.53–0.54	0/10
RHD deaths, all ages	0.64–0.84	3/10	3.21–4.09	0/10
RHD deaths, 0–19	0.98–1.39	7/10	3.23–3.98	0/10
RHD deaths, 20+	0.57–0.74	0/10	3.20–4.11	0/10
Prevalence, 0–19 (diagnostic)	0.86–0.86	3/10	0.64–0.75	0/10
Incidence (validation)	0.82–0.82	–	0.35–0.35	–

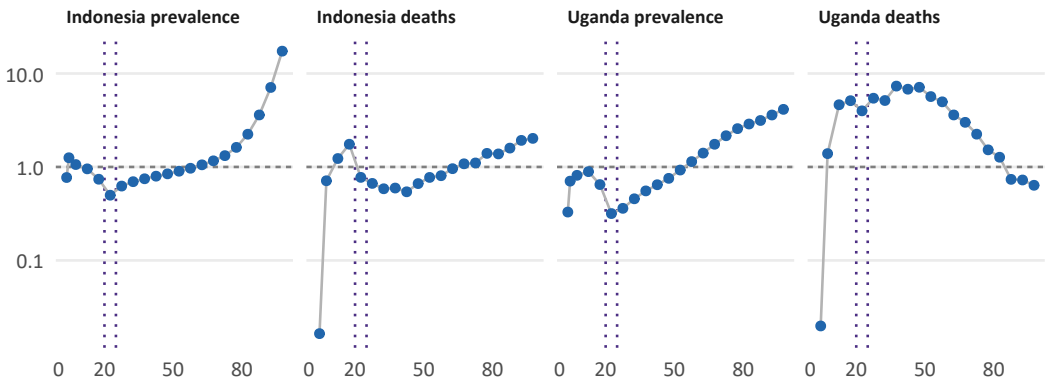
Ratios over the scored years, both sexes, GBD rate $\times$ WPP population. Tolerance $|\log \text{ratio}| < 0.15$ (prevalence) and 0.25 (deaths) in every year. Ratios to published GBD counts are in backup.

Over 2010–2019 the death ratios rise while GBD deaths per prevalent case fall



Model / GBD target by year, log scale. Green band: tolerance; dotted line: start of the scored years. GBD deaths per prevalent case change -21% (Indonesia) and -13% (Uganda) from 2010 to 2019.

By age, prevalence dips at 20–24, where prior-A leaves the count, and adult deaths are above GBD



Model / GBD target by GBD age group, 2019, both sexes, log scale. Purple dotted lines: ages 20–24. Prevalence at 20–24 is 0.50 (Indonesia) and 0.32 (Uganda) times GBD.

Calibrated transitions hit their bounds: A → normalised, A → B, B → C, C → D (Indonesia); A → normalised, A → B, B → A, B → C, C → B, C → D (Uganda)

Annual probability	Centre [bounds]	Indonesia	Uganda
A → normalised	0.125 [0.007, 0.25]	**0.2490**	**0.0800**
A → B	0.054 [0.002, 0.07]	**0.0723**	**0.0300**
B → A	0.177 [0.004, 0.20]	0.0510	**0.0800**
B → C	0.074 [0.001, 0.13]	**0.1253**	**0.0800**
C → B	0.050 [0.012, 0.08]	0.0342	**0.0250**
C → D	0.112 [0.006, 0.25]	**0.2497**	**0.0800**
B → C, age >20 (derived)	hazard × 0.10 (IDN) / 0.50 (UGA)	0.0133	0.0408
C → D, age >20 (derived)	hazard × 0.10 (IDN) / 0.50 (UGA)	0.0283	0.0408

Amber bold: at a bound. Incidence multipliers: 0.63–1.19 (Indonesia), 0.32–0.87 (Uganda). Calibrated rows are childhood values. Fixed: D death, adult B normalisation, adult progression multiplier 0.10 (IDN) / 0.50 (UGA), entry mixes, D→C = 0.

Stage D holds 10% of Indonesia's modelled RHD against a GBD heart-failure proxy of 3.2%

	Indonesia	Uganda
Stage-D annual RHD death probability, age <20 / 20+ (fixed input)	16.0% / 4.0%	6.0% / 3.0%
Realised D death risk, all ages, 2019	4.8%	3.1%
Stage-D share of classifiable RHD, 2019	9.5%	11.0%
GBD heart-failure proxy share (diagnostic)	3.2%	0.9%
Model / GBD RHD deaths per prevalent case, 2019	1.1	6.8

D mortality depends only on attained age and is the same for every disease and diagnosis history (the former late-entry multiplier of 2 is removed). Realised risk is lower because of competing other-cause death and reference surgery; the all-age value is dominated by adult D.

Four mechanisms limit the fit under the fixed assumptions

Mechanism	Evidence	Why it binds
Adult progression tied to childhood rates	GBD child / adult case fatality 0.63% / 0.41% (Indonesia)	Childhood deaths need fast progression; adults inherit it at 0.10 (IDN) / 0.50 (UGA)$\times$ the hazard and still over-progress
Prior-A exclusion at age 21	Prevalence 20–24: 0.50 / 0.32 $\times$ GBD	Child A counts as RHD; the same lesion stops counting at 21
Too few adult exits	Uganda prevalence 60+: 1.75 $\times$ GBD	Adult B leaves only via 2% normalisation or progression
Falling GBD case fatality	Deaths per case -21% / -13%, 2010–2019	Constant natural history cannot reproduce a declining trend

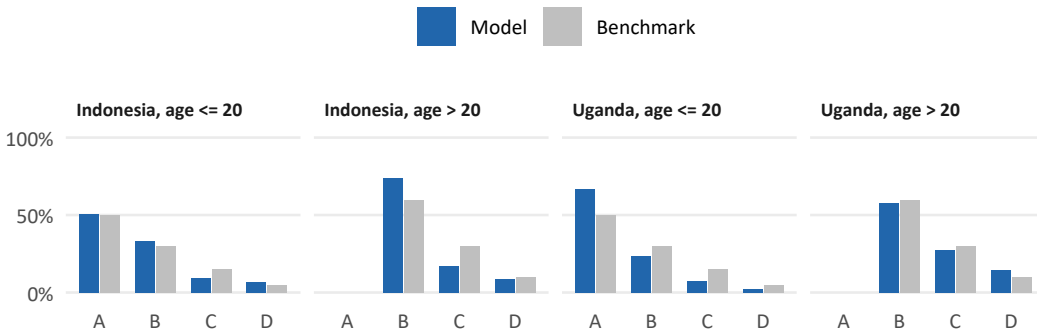
Evidence from the saved Layer-2 targets and fit (Indonesia / Uganda where two values are shown).

In the 5 Nov scratch tests no variant passed; adult parameters plus prior-A in prevalence came closest

Variant	IDN prevalence (all / 20+)	IDN deaths (all / 0–19 / 20+)	UGA prevalence (all / 20+)	UGA deaths (all / 0–19 / 20+)	Pass?
Shared rates (5 Nov run)	0.86–0.87 / 0.83–0.84	2.9–3.9 / 0.36–0.50 / 3.5–4.5	0.74–0.78 / 0.71–0.73	1.5–1.9 / 0.58–0.71 / 1.6–2.1	no
+ calibrated adult B to C, C to D	0.75–0.76 / 0.75	1.7–2.2 / 0.89–1.26 / 1.8–2.4	—	—	no
+ calibrated adult B to normalised	—	—	0.87–0.91 / 0.83–0.85	1.5–1.9 / 0.35–0.43 / 1.6–2.1	no
+ both (3 adult parameters)	0.86–0.87 / 0.85–0.86	1.14–1.48 / 0.89–1.26 / 1.19–1.52	0.82–0.86 / 0.76–0.78	1.04–1.33 / 0.84–1.03 / 1.07–1.37	no
+ both, prior-A counted in prevalence	0.87–0.89 / 0.87–0.88 (in tol.)	1.12–1.46 / 0.86–1.22 / 1.18–1.51	0.88–0.92 / 0.86–0.88 (in tol.) (20+ marginally out in 2010: 0.860)	1.03–1.31 / 0.84–1.03 / 1.06–1.35	no

Scratch tests on the previous shared-rate model (dev/ checkpoint note; not saved outputs). The current model fixes the adult multiplier at 0.10 (IDN) / 0.50 (UGA).

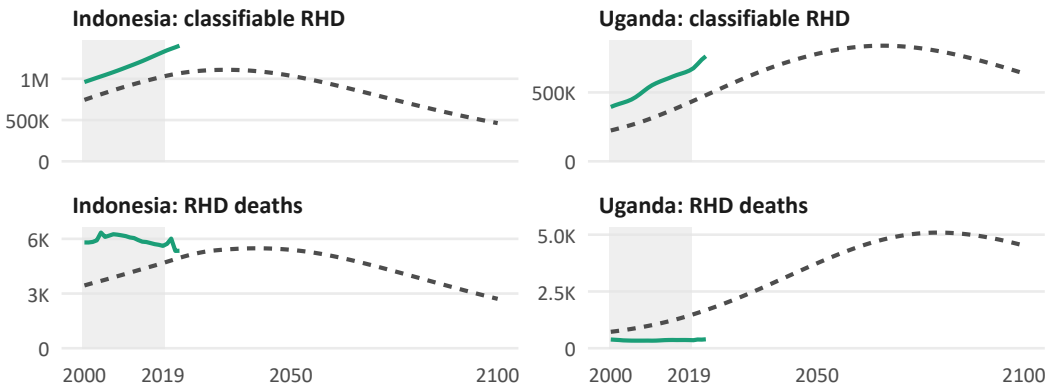
In 2010–2019, 74% of Indonesia's adult first diagnoses are stage B, against a provisional benchmark of 60%



Stage at first confirmed diagnosis (screen + clinical), reference care. Benchmark: provisional illustrative mixes, a diagnostic comparison only — not a fit term or a national stage target.

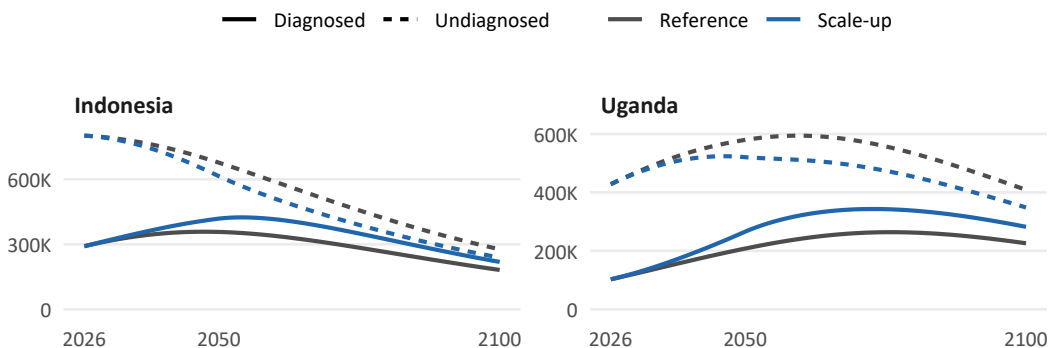
What the preliminary model projects

Projected 2026 RHD deaths are 1.0× GBD 2023 in Indonesia and 4.8× in Uganda



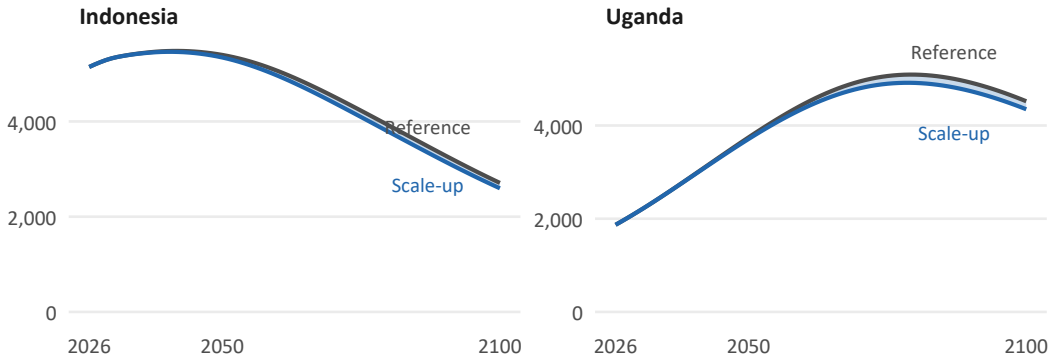
Green: GBD 2023 published counts. Dashed grey: model (calibration window 2000–2019, shaded; bridge; reference arm 2026–2100). All ages, both sexes.

Scale-up raises the diagnosed share of classifiable RHD to 41% in Indonesia and 34% in Uganda by 2050



Classifiable RHD at the end of each year: diagnosed (solid) and undiagnosed (dashed); grey reference, blue scale-up. Total classifiable RHD is nearly identical across arms.

Scale-up averts about 6,300 RHD deaths in Indonesia and 7,400 in Uganda by 2100, up to 2.4% of reference-arm deaths



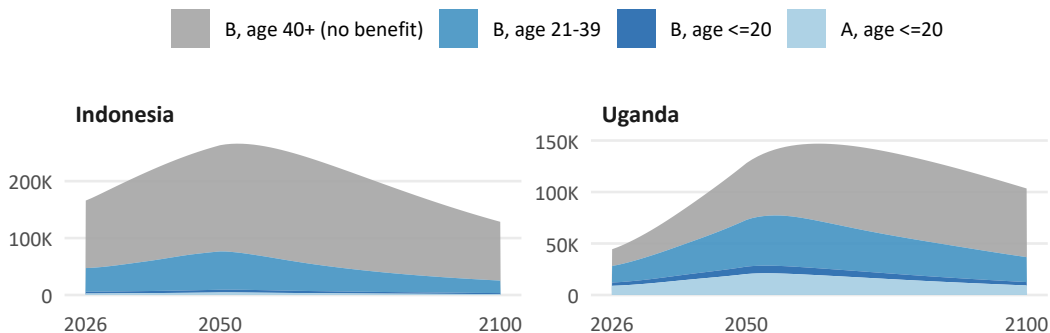
Annual RHD deaths. Cumulative reduction: 1.84% (Indonesia) and 2.41% (Uganda) of reference-arm deaths, 2026–2100; 400 and 228 of the totals occur by 2050.

By 2050 scale-up delivers 4.9 million screens a year in Indonesia but adds only 7,344 screen-detected first diagnoses

	Indonesia		Uganda	
Per year, 2050	Reference	Scale-up	Reference	Scale-up
Screening encounters	0.21 m	**4.91 m**	0.02 m	**1.80 m**
True screen positives	394	**8,554**	82	**9,310**
Confirmations completed (uncosted)	355	**7,699**	74	**8,379**
First diagnoses via screening	355	**7,699**	74	**8,379**
First diagnoses, clinical	13,619	**12,307**	11,399	**10,155**
Diagnosed classifiable RHD (end)	356,965	**418,645**	208,061	**266,058**
People on BPG (person-years)	190,295	**263,162**	75,745	**128,159**
Valve surgeries	1,554	**1,541**	2,106	**2,081**

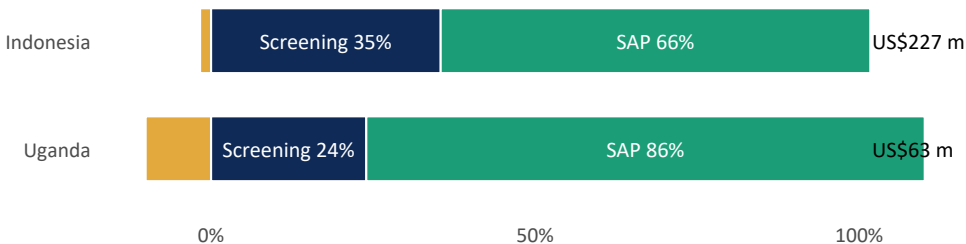
Counts from the engine's care trace; m = million. Encounters include people without RHD. On-BPG person-years: confirmed A (≤ 20) and B.

71% (Indonesia) and 43% (Uganda) of scale-up BPG person-years in 2050 are stage B at age 40 or over, costed with no modelled benefit



Scale-up arm, people on BPG by stage and attained age (person-years). Every on-BPG person-year is costed; the assumed progression benefit applies only at ages ≤ 39. The adherence-weighted exposure (on BPG $\times$ a) counts every age: 71% (Indonesia) and 43% (Uganda) of it in 2050 is at ages ≥ 40, where it confers no modelled benefit.

BPG person-years are 66% of Indonesia's and 86% of Uganda's discounted incremental cost



Discounted (3%, base 2026) incremental cost, constant 2025 US$, 2026–2100, by component; screening 35% (Indonesia) and 24% (Uganda). Confirmatory echoes (239,655 Indonesia, 306,236 Uganda) are not priced.

Benefit–cost ratios are 18.78 in Indonesia and 14.22 in Uganda over 2026–2100 — on an inadequate baseline

Discounted to 2026 at 3%, constant 2025 US$	Indonesia	Uganda
Incremental cost	227.1 m	63.5 m
Monetised benefit	4,266.6 m	902.6 m
Net benefit	4,039.4 m	839.2 m
Benefit–cost ratio	**18.78**	**14.22**
Benefit–cost ratio, undiscounted	33.59	26.41
Net benefit per resident in 2026	US$13.98	US$15.70
RHD deaths averted (undiscounted)	6,300	7,386
Cost per death / per DALY averted	US$36,050 / US$4,598	US$8,592 / US$1,240
Confirmatory echoes not costed (incremental)	239,655	306,236

Benefit = deaths averted × VSL only (reference case: PPP income elasticity 1.5, floor 20× GNI per capita; 2026 VSL US$373,000 and US$47,000). Per resident: discounted net benefit / 2026 population. m = million. Deterministic; sensitivity analysis pending; costs understated by the confirmation cost.

Ending the horizon in 2050 lowers the ratios to 2.18 and 0.71

	Indonesia		Uganda	
	2026–2050	2026–2100	2026–2050	2026–2100
RHD deaths averted	425	6,300	228	7,400
Benefit–cost ratio	**2.18**	**18.78**	**0.71**	**14.22**
Net benefit (discounted, 2025 US$)	107.3 m	4,039.4 m	-6.2 m	839.2 m

2026–2050 columns are deck-only sums of the saved runs to 2050; the model runs forward only, so this equals a 2050 run with the same inputs.

What must be resolved

The engine and pipeline run end to end; the calibration and key assumptions are not settled

Implemented

- First-diagnosis engine
- Adult hazard multiplier (0.10 (IDN) / 0.50 (UGA))
- Age-specific D mortality (< 20 / 20+)
- Screening, confirmation, BPG cascade
- Joint calibration; bridge
- Outputs, costing, BCR

Not settled

- Fit to GBD (fails)
- Adult progression (fixed, not estimated)
- Prior-A prevalence mapping
- SAP progression effect (assumed), protection age
- Confirmation unit cost

Planned

- Recalibration decision
- Sensitivity analyses
- Report rewrite
- Adherence evidence

A credible fit still needs a decision on a fixed assumption; the adopted adult multiplier is a partial step

Option	Change	Evidence
1 (recommended in the 5 Nov checkpoint)	Free adult B→C, C→D, B→normalised; count adult prior-A as prevalence	Prevalence within tolerance; deaths 1.0–1.5×, drifting up
2	Free the three adult parameters only; keep prior-A excluded	Deaths 1.0–1.5×; prevalence 0.76–0.87×
3 (current model, adopted 6 Nov)	Fix the adult hazard multiplier (0.10 (IDN) / 0.50 (UGA)) on B→C and C→D; fixed age-specific D mortality (9 Nov); keep prior-A excluded	Deaths 0.64–0.84× (Indonesia), 3.21–4.09× (Uganda); prevalence 0.78–0.79 / 0.56–0.60×

Options 1–2: scratch tests on the previous shared-rate specification (dev/checkpoint_first_diagnosis_2026-11-05.md). Option 3: the saved production runs in this deck (scored 2010–2019). Passing the death tolerance in every scored year would additionally need a historical case-fatality or care trend, or a different scoring rule.

Remaining limitations bias results in directions we can name

Limitation	Likely effect on results	What resolves it
Baseline deaths	Benefit per death unclear	Recalibration decision
Assumed SAP effect	Size unknown; B 40+ costed, no benefit	Progression evidence; benefit age
Confirmation cost	Cost understated	Unit cost for confirmatory echo
Initial care shares	Shape diagnosed share	Programme data
Transferability	Direction unknown	Country cohort or registry data
Uncertainty	Spread unknown	Probabilistic, one-way analyses

On preliminary runs, scale-up returns US$18.8 (Indonesia) and US$14.2 (Uganda) per dollar through 2100 — but the calibration fails and the SAP progression effect is assumed, not estimated, so these numbers are not yet usable.

Five collaborator decisions will define the next model version

1. Should the fixed adult hazard multiplier (0.10 (IDN) / 0.50 (UGA)) give way to adult rates estimated from data, and should adult normalisation be estimated?
2. Should adult prior-A count as RHD prevalence, to match the GBD case definition across age 21?
3. Should the calibration reproduce GBD's falling case fatality, or score fewer years?
4. Is the assumed 55% progression-hazard reduction to age 39 acceptable, and should stage B at 40+ stay on BPG?
5. What does a confirmatory echo cost, and what completion rate is realistic?

Backup slides hold detail for questions and sit outside the 40-minute budget

- Calibration objective, cells, target basis, fit by sex
- Incidence multipliers, initialisation, convergence
- Identifiability; literature transitions; stage-D proxies
- Care trajectories by arm
- Economic parameters, detail, budget impact
- Abstract; sensitivity; discrepancies; provenance

Layer 2 minimises normalised log errors with child and adult components plus weak priors

$$L_2 = w_P\left[p\,\bar{\ell}_{P,\text{0-19}} + (1-p)\,\bar{\ell}_{P,20+}\right] + w_D\left[s\,\bar{\ell}_{D,\text{0-19}} + (1-s)\,\bar{\ell}_{D,20+}\right] + w_I\bar{\ell}_I + \lambda\Big[\sum_k \frac{(\text{logit}\,p_k - \text{logit}\,p_k^0)^2}{\sigma_p^2} + \sum_{b,s}\frac{(\ln\alpha_{bs})^2}{\sigma_\alpha^2}\Big] + \lambda_\Delta\sum_s\sum_b(\Delta^2\ln\alpha_{bs})^2$$

$\bar{\ell}$: weighted mean of $[\ln(X^{\text{mod}}+1) - \ln(X^{\text{tgt}}+1)]^2$ over unmasked cells, both sexes pooled. w_P, w_D, w_I = 2.5, 1.0, 0.0; p = 0.5, s = 0.5; λ = 0.05, λ_Δ = 0.5; σ_p = 2, σ_α = 1. Optimum L = 2.517 (Indonesia), 3.313 (Uganda); prior / data 0.08 and 0.20.

Childhood deaths are the largest weighted term in Indonesia's objective

Term at the optimum	Indonesia	Uganda
Prevalence 0–19 (raw)	0.046	0.361
Prevalence 20+ (raw)	0.844	0.757
Deaths 0–19 (raw)	2.237	1.200
Deaths 20+ (raw)	0.192	1.526
Prevalence (weighted)	1.112	1.398
Deaths (weighted)	1.214	1.363
Prior: transitions	0.052	0.068
Prior: incidence α	0.070	0.429
Smoothness: α	0.069	0.055

Cells: prevalence 100 + 320, deaths 100 + 315 (Indonesia); deaths 80 + 310 (Uganda).

Layer-2 targets use GBD rates times WPP population, which differ from published counts by up to 10%

	Indonesia		Uganda	
Measure	Range	Window	Range	Window
Deaths	0.958–1.020	0.973	0.977–1.084	1.031
Incidence	0.964–1.033	0.997	0.933–1.013	0.981
Prevalence	0.966–1.025	0.983	0.904–1.014	0.974

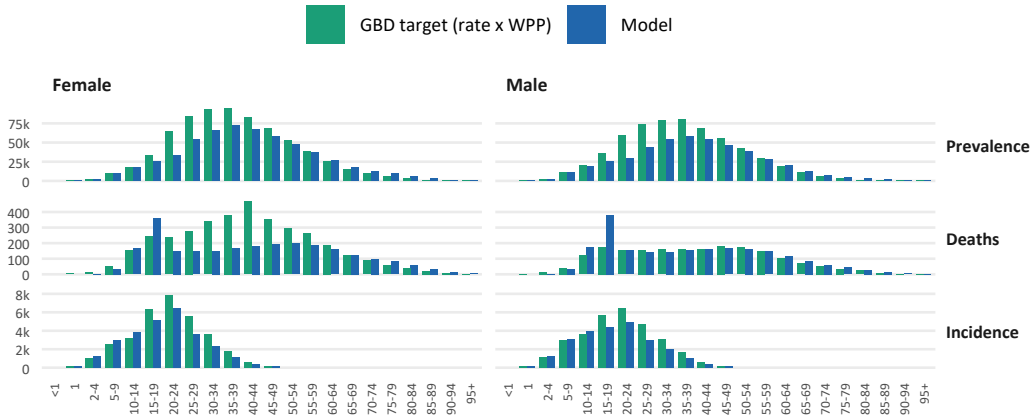
Ratio of GBD published Number to GBD Rate × WPP population, by sex and year, 2000–2019. Model / GBD Number ratios for the checkpoint quantities are on the next slide.

Against published GBD counts the fit is similar

	Indonesia		Uganda	
2019: target / Number	Female	Male	Female	Male
Prevalence All ages	0.81 / 0.80	0.77 / 0.75	0.64 / 0.67	0.56 / 0.62
Prevalence 20+	0.80 / 0.79	0.76 / 0.74	0.58 / 0.61	0.50 / 0.57
Deaths All ages	0.69 / 0.69	1.12 / 1.09	3.42 / 3.35	5.24 / 5.10
Deaths 0–19	1.19 / 1.18	1.66 / 1.64	3.62 / 3.87	4.39 / 4.58
Deaths 20+	0.62 / 0.62	1.00 / 0.98	3.40 / 3.30	5.40 / 5.19

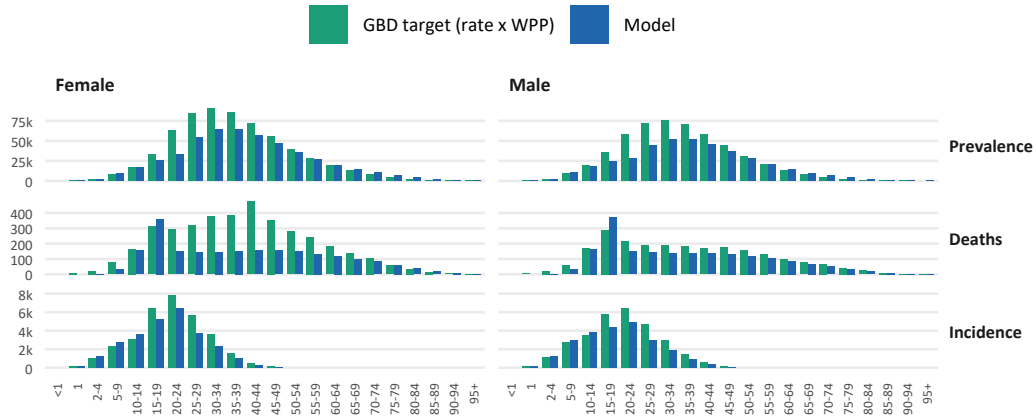
Model / (GBD rate × WPP) and model / GBD published Number. Uganda's WPP population exceeds GBD's, so its rate-based targets are higher than published counts.

Indonesia, 2019: model / GBD is 0.79 for prevalent cases, 0.84 for deaths and 0.82 for incidence



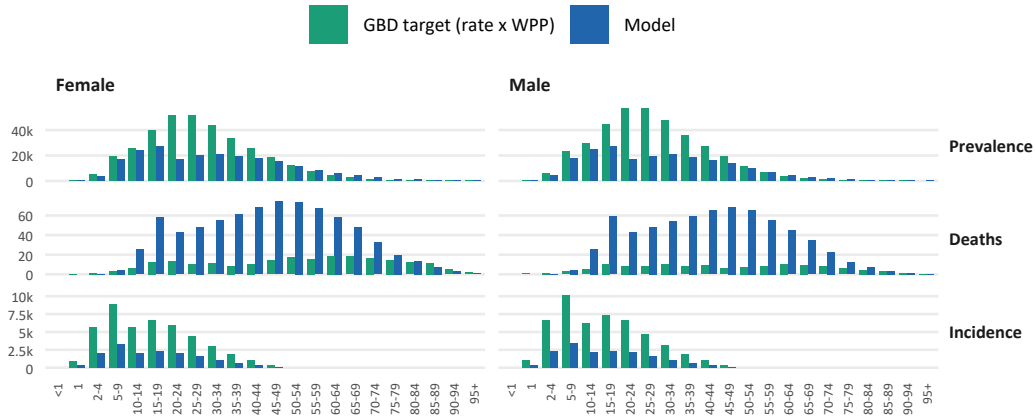
Counts per year by GBD age group and sex; GBD = GBD rate × WPP population. Own y scale per panel. Incidence = new modelled disease (validation only).

Indonesia, 2000–2019 (annual mean): model / GBD is 0.78 for prevalent cases, 0.66 for deaths and 0.82 for incidence



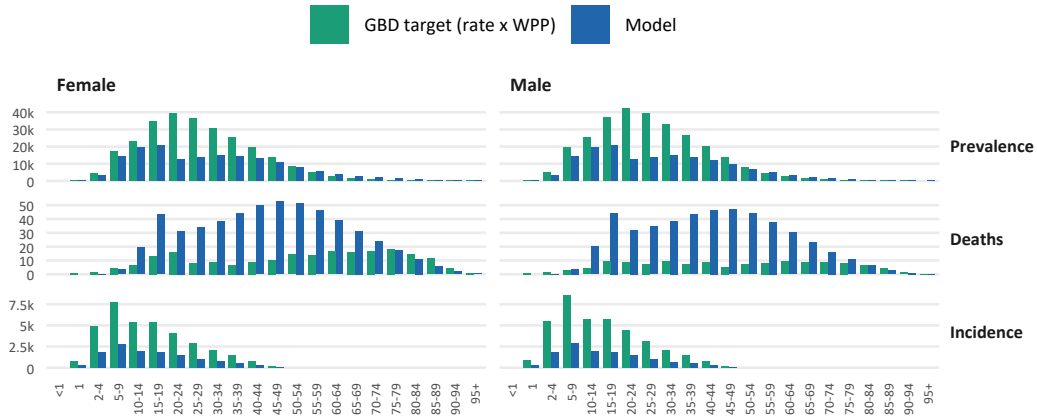
Counts per year by GBD age group and sex; GBD = GBD rate × WPP population. Own y scale per panel. Incidence = new modelled disease (validation only). Mean of 2000–2019; 2010–2019 scored.

Uganda, 2019: model / GBD is 0.60 for prevalent cases, 4.09 for deaths and 0.35 for incidence



Counts per year by GBD age group and sex; GBD = GBD rate × WPP population. Own y scale per panel. Incidence = new modelled disease (validation only).

Uganda, 2000–2019 (annual mean): model / GBD is 0.58 for prevalent cases, 3.02 for deaths and 0.35 for incidence



Counts per year by GBD age group and sex; GBD = GBD rate × WPP population. Own y scale per panel. Incidence = new modelled disease (validation only). Mean of 2000–2019; 2010–2019 scored.

Layer-2 incidence multipliers move away from the Layer-1 start values

	Indonesia		Uganda	
Age band	Layer-1 start	Layer 2 (used)	Layer-1 start	Layer 2 (used)
0-14	0.98 / 0.97	1.19 / 1.07	0.91 / 0.91	0.37 / 0.34
15-24	0.98 / 0.97	0.82 / 0.76	0.94 / 0.93	0.35 / 0.32
25-44	0.98 / 0.98	0.65 / 0.63	0.95 / 0.95	0.37 / 0.33
45-64	0.99 / 0.99	0.64 / 0.63	0.98 / 0.98	0.47 / 0.43
65-79	1.00 / 1.00	0.68 / 0.69	1.00 / 0.99	0.63 / 0.60
80+	1.00 / 1.00	0.75 / 0.77	1.00 / 1.01	0.87 / 0.85

Female / male multipliers on the GBD incidence pattern (per total population). Incidence is validation in Layer 2: model / GBD 0.82 (Indonesia), 0.35 (Uganda) in 2019.

The burn-in converges and the bridge carries the full state to 2026 without reseeding

	Indonesia	Uganda
Burn-in cycles / doubled check	120 / 240	120 / 240
Largest change, doubled burn-in	3.0e-11	1.1e-10
Bridge prevalence / GBD, 2023	0.79	0.57
Bridge RHD deaths / GBD, 2023	0.95	4.26
Seed entering 2026 / GBD-rate prevalence	0.78	0.54
Calibration → bridge hand-off gap	0.0e+00	0.0e+00

Zero-start burn-in under 2000 inputs with per-capita balancing; bridge 2020–2025 under reference care. The seed keeps diagnosis, BPG, prior-A, normalised and screening histories.

All optimiser starts converge; the starts reach 2 (Indonesia) and 1 (Uganda) distinct optima

Country	Start	Objective	Code	Evaluations
Indonesia	prior center	2.5174	0	41
Indonesia	layer1 alpha	2.5174	0	42
Indonesia	prevalence first	2.5174	0	46
Indonesia	random 1	2.5174	0	48
Indonesia	random 2	2.5959	0	55
Uganda	prior center	3.3132	0	40
Uganda	layer1 alpha	3.3132	0	39
Uganda	prevalence first	3.3132	0	36
Uganda	random 1	3.3132	0	45
Uganda	random 2	3.3132	0	38

Bounded L-BFGS-B from the prior centre, Layer-1 values, a prevalence-first start and 2 random starts. Convergence code 0 is not fit adequacy.

A one-at-a-time sweep flags 2 (Indonesia) and 4 (Uganda) parameters as prior-dominated

Parameter	Indonesia	Uganda
A $\rightarrow$ normalised	0.27	**0.82**
A $\rightarrow$ B	0.36	**0.73**
B $\rightarrow$ A	**0.82**	**0.55**
B $\rightarrow$ C	0.09	0.27
C $\rightarrow$ B	**1.00**	**1.00**
C $\rightarrow$ D	0.09	0.18
Incidence α, 0-14 (F / M)	0.18 / 0.18	0.18 / 0.18
Incidence α, 15-24 (F / M)	0.18 / 0.09	0.18 / 0.18
Incidence α, 25-44 (F / M)	0.09 / 0.18	0.09 / 0.18
Incidence α, 45-64 (F / M)	0.18 / 0.18	0.18 / 0.18
Incidence α, 65-79 (F / M)	0.09 / 0.09	0.18 / 0.18
Incidence α, 80+ (F / M)	0.27 / 0.27	0.36 / 0.36

Flatness: share of a parameter's range within the near-optimal band, others at the optimum; amber = prior-dominated. Weaker than joint identification.

Care trajectories differ only in screening and BPG; surgery is identical in both arms

	Indonesia		Uganda	
Care input	Reference	Scale-up	Reference	Scale-up
One screen across ages 5–39	5%	5% → 80%	1%	1% → 80%
Clinical detection (annual)	2%	2% → 2%	2%	2% → 2%
BPG initiation (annual)	50%	50% → 90%	50%	50% → 90%
BPG retention (annual)	90%	90% → 95%	90%	90% → 95%
Good adherence share	60%	60% → 85%	60%	60% → 85%
Partial adherence share	20%	20% → 10%	20%	20% → 10%
Surgery coverage	5%	5% → 5%	3%	3% → 3%

Scale-up ramps linearly 2026–2050, then holds. The reference arm equals the care applied in calibration (checked at render).

Published annual transition probabilities vary 7- to 108-fold across studies

Transition	Studies	Minimum	Maximum	Indonesia	Uganda
A → normalised	11	0.0074	0.2490	0.2490	0.0800
A → B	15	0.0015	0.0723	0.0723	0.0300
B → A	11	0.0042	0.2025	0.0510	0.0800
B → C	9	0.0012	0.1253	0.1253	0.0800
C → B	4	0.0119	0.0810	0.0342	0.0250
C → D	4	0.0057	0.2497	0.2497	0.0800
D → C	3	0.0042	0.0710	0.0000	0.0000

Table 2 of the collaborators' September 2026 worksheet, annualised. Model values are the calibrated or fixed values in force (childhood values; adult B→C and C→D = hazard × 0.10 (IDN) / 0.50 (UGA): Indonesia 0.0133 / 0.0283, Uganda 0.0408 / 0.0408).

Stage-D mortality depends on its proxy: the two definitions differ up to 19-fold

		Worksheet stage-D death rate	
	Model input (fixed)	Literature proxy	GBD heart-failure proxy
Indonesia	%	2.4%	11.9%
Uganda	%	0.3%	6.0%

Worksheet rates divide GBD 2023 RHD deaths by an estimated stage-D count; collaborator worksheet, October 2026. All ages, both sexes.

Economic parameters differ by country in income, growth and unit costs

Parameter	Indonesia	Uganda
Discount rate	3%	3%
Discount base year	2026	2026
GNI per capita, PPP (VSL income anchor)	int$11,620	int$2,380
Income anchor year	2019	2019
Unfloored VSL / GNI per capita at the anchor	70.0×	31.7×
VSL in 2026, constant 2025 US$	US$373,000	US$47,000
VSL in 2100, constant 2025 US$	US$7,987,000	US$1,544,000
Real income growth (GDP per capita proxy)	2.8%	3.2%
DALYs per RHD death	30	30
Price factor, assumed 2019 to 2025	1.000	1.312
Cost per screening encounter	US$1.10	US$0.52
Cost per person-year on BPG	US$110	US$52
Cost per surgery	US$8,998	US$6,560
Cost per confirmatory echo	not costed	not costed
International $ per US$, 2025	3.490	2.834

VSL: U.S. US$9.4 m and int$57,900 (2015); elasticity 1.5, floor 20× GNI per capita. Macro data: World Bank WDI, U.S. CPI-U. DALY: deaths only. Sensitivity analysis pending.

Annual incremental cost peaks in 2052 (Indonesia) and 2057 (Uganda); the scale-up ramp ends in 2050

	Indonesia, 2025 US$ m				Uganda, 2025 US$ m			
Year	Ref.	Scale-up	Incr.	Confirm.	Ref.	Scale-up	Incr.	Confirm.
2030	32.7 m	34.2 m	1.5 m	1,290	10.7 m	10.9 m	0.3 m	1,002
2040	35.2 m	42.0 m	6.8 m	4,517	14.3 m	15.9 m	1.6 m	4,232
2050	35.2 m	48.2 m	13.1 m	7,344	17.8 m	21.3 m	3.5 m	8,305
2060	32.9 m	45.3 m	12.4 m	4,169	20.6 m	24.4 m	3.8 m	5,775
2070	29.4 m	40.1 m	10.7 m	2,783	22.2 m	25.6 m	3.4 m	4,339
2080	25.3 m	34.3 m	9.0 m	2,113	22.4 m	25.3 m	2.9 m	3,512
2090	21.2 m	28.8 m	7.6 m	1,696	21.3 m	23.8 m	2.5 m	2,894
2100	17.5 m	23.9 m	6.4 m	1,369	19.3 m	21.5 m	2.2 m	2,339

Undiscounted annual programme cost by arm (screening + BPG + surgery), selected years; Confirm. = incremental confirmatory echoes, a volume, not priced.

The submitted abstract used earlier runs and a different specification; the preliminary runs do not reproduce it

	Uganda (low-income illustration)		Indonesia (upper-middle-income)	
	Abstract	Saved run	Abstract	Saved run
RHD deaths averted by 2050	7,900	228	23,000	425
Death-rate reduction in 2050	30%	0.9%	24%	1.0%
Benefit–cost ratio	1.28	0.71	1.42	2.18

Abstract values parsed from the submitted WCRHD 2026 abstract (hypothetical countries, earlier runs). Saved-run values are sums to 2050 of the current preliminary outputs.

No sensitivity analysis exists yet; these analyses are planned

- **Structural:** adult progression multiplier (now fixed at 0.10 (IDN) / 0.50 (UGA)) and adult normalisation; prior-A in prevalence; direct advanced entry in children
- **Calibration:** prevalence : death weights; death-trend handling; target basis
- **One-way:** assumed SAP progression multiplier M and protection age (GOAL-trial child hazard ratios as the documented alternative); BPG initiation, retention, adherence; confirmation completion; screening window target
- **Economic:** confirmation unit cost; VSL reference value, income elasticity and floor; income growth; discount rate; unit costs and their price years
- **Probabilistic:** transitions and incidence multipliers within bounds

The pipeline is deterministic; none of the results in this deck carry uncertainty intervals.

Six discrepancies between documents, configuration and code are logged

Item	Detail	Action
Indonesia incidence trend	Saved run 0.985; scripts/00_run_all.R 0.985 (consistent)	None
Pipeline control	PIPELINE_RUN_THROUGH = 08: 05–08 run under a user override of the failed fit	Recalibrate
Project guide	Consistent	None
Engine tests	tests/ target an earlier engine signature and output columns	Update tests
State diagram file	docs/ PNG predates the first-diagnosis structure; the deck draws its own	Redraw
Confirmation cost	Volume tracked in 06 and 08; no unit cost in 00 block J	Decide

Provenance (1/2): calibration and health results trace to saved objects

Result	Script	Artifact	Object	Shape	Transform
Fit, statuses	04b	data/<C>/calibrated_rhd_parameters.rds	$stage_calibration$fit_tolerance_summary, $fit_by_year, $statuses	years × quantities	ranges
Parameters, bounds	04b	same	$stage_calibration$parameters, $bounds	45 rows	none
Mechanisms	04b	same	$fit_by_year targets, $fit_by_group_year	age group × year	ratios
Validation	01, 04b, 06	data/<C>/temp_baseline_rates_gbd.rds; same; output/<C>/out_model/<C>.rds	GBD Number; $fit_by_year; $wsd, $stages (ref)	year	sums
Deaths averted	06	output/<C>/out_model/<C>.rds	wsddead	101 ages × 2 sexes × 75 years × 2 arms	ref − sap; cumulative
Care states, volumes	06	same	$stages undiagnosed_end, diagnosed_end, n_dx_*, n_on_bpg_*	same; persons	sums by arm and year

Provenance (2/2): service, cost and economic results trace to saved objects

Result	Script	Artifact	Object	Shape	Transform
Care inputs, seed	05	data/<C>/baseline_state.rds	$states[[1]]$coverage, $seed_state	vectors of 75 years	none
Service volumes	06	output/<C>/out_model/<C>.rds	$stages n_screened, n_confirm_completed, n_on_optimal_treatment	persons per year	sum by arm and year
Cost composition	06, 08	same + rhd_economic_results.rds	$stages volumes × $params unit costs	2025 US$	× discount factor; sap — ref
Economic summary	08	output/<C>/tables/rhd_economic_results.rds	$summary, $incremental	1 row per location	none
Uncosted confirmation	06, 08	same	$budget_impact, $params	per year	sums
Literature, stage-D proxies	none	dev/*.xlsx	Table 2; All-ages rows	probabilities	min, max
Alternatives tested	none	dev/checkpoint_first_diagnosis_2026-11-05.md	Section 10 table	text	parsed

<C> = Indonesia or Uganda; paths relative to the repository root. The deck reads these files and writes nothing.

Rendering needs only the saved artifacts, R and a XeLaTeX toolchain

- Render: `rmarkdown::render("reports/slideck-technical.Rmd")` with XeLaTeX and Calibri; set `RHD_REPO_ROOT` when rendering a copy outside the repository
- Inputs: artifacts of scripts 01–08 for both countries; a missing file stops the render and names its producing script
- Checks: engine and specification, seed = bridged state, reference care = reference arm, care trajectories, SAP trace, nesting, volumes and economics agree for each country
- Software: R version 4.5.1 (2025-06-13 ucrt); rmarkdown 2.29; knitr 1.50; data.table 1.17.8; ggplot2 4.0.3; kableExtra 1.4.0
- Saved: calibration 2026-11-10 / 2026-11-09; model outputs 2026-11-10 / 2026-11-09 (Indonesia / Uganda)

References

Coates, M. M., Sliwa, K., Watkins, D. A., Zühlke, L., Perel, P., Berteletti, F., Eiselé, J. L., Klassen, S. L., Kwan, G. F., Mocumbi, A. O., Prabhakaran, D., Habtemariam, M. K., & Bukhman, G. (2021). An investment case for the prevention and management of rheumatic heart disease in the african union 2021–30: A modelling study. *The Lancet Global Health*, *9*, e957–e966. https://doi.org/10.1016/S2214-109X(21)00199-6

Rwebembera, J., Marangou, J., Mwita, J. C., Mocumbi, A. O., Mota, C., Okello, E., Nascimento, B., Thorup, L., Beaton, A., Kado, J., Kaethner, A., Kumar, R. K., Lawrenson, J., Marijon, E., Mirabel, M., Nunes, M. C. P., Piñeiro, D., Pinto, F., Ralston, K., … Reményi, B. (2023). 2023 world heart federation guidelines for the echocardiographic diagnosis of rheumatic heart disease. *Nature Reviews Cardiology 2023 21:4*, *21*, 250–263. https://doi.org/10.1038/s41569-023-00940-9